

"TABANG CORDOVA: A Geolocation-Based Emergency and Assistance Coordination System"

System Overview

Target Users

Primary Users:

- Registered Barangay Residents of Cordovas

Responders:

- Barangay Officials and Utilities
 - Municipal Authorities such as the Philippine National Police
 - Bureau of Fire Protection
 - Municipal Disaster Risk Reduction and Management Office (MDRRMO)
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User Needs

Residents need:

- Immediate way to request help
- Fast response during emergencies
- Clear communication with authorities
- Assurance that their request is seen and acknowledged

Barangay and Municipal authorities need:

- Real-time visibility of incidents
 - Accurate location tracking
 - Organized monitoring system
 - Clear escalation process for serious cases
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Current Situation

Currently, residents rely on:

- Direct phone calls
- Facebook posts
- Personal reporting to barangay hall

- Word of mouth

Problems with this setup:

- Delayed response
- Miscommunication of location
- No centralized monitoring
- Difficult coordination between barangay and municipal level
- Lack of documentation or tracking history

In critical incidents such as fire or armed quarrels, barangay officials often struggle to quickly relay accurate information to higher authorities.

Proposed Solution

The system operates through a two-level response structure: **Barangay Level and Municipal Level.**

◆ Resident Flow

- Resident logs into the system (account stays logged in even if the browser is closed).
- On the homepage, they see a live map of their barangay.
- They click the “Request Response” button.
- They upload a photo as proof of the situation.
- The system automatically captures their location.
- The request is submitted instantly.

◆ Barangay & Municipal Monitoring

On the live monitoring dashboard:

- Each request appears as a **pinpoint location on the map.**
- When officials hover over the pin, they can see:
 - Photo of the situation
 - Name of the resident
 - Contact number
 - Barangay cluster classification

Barangay officials act as primary responders.

If the case involves:

- Fire outbreaks
- Armed quarrels
- Severe medical emergencies
- Large-scale disasters

The request can immediately be coordinated with municipal authorities such as:

- Philippine National Police
- Bureau of Fire Protection
- MDRMO

This ensures proper escalation for incidents beyond barangay capacity.

Differentiator

- ✓ Real-time map-based monitoring system
- ✓ Direct resident-to-authority communication
- ✓ Photo-based verification to prevent fraudulent requests
- ✓ Cluster-based barangay monitoring
- ✓ Unified dashboard for both barangay and municipal level
- ✓ Continuous system — not only for disaster events

AI Feature

Heatmap & Risk Prediction Feature

The system includes a Heatmap and Risk Prediction module that analyzes past incident reports based on location, type of incident, and time of occurrence.

It generates:

- High-risk barangay clusters
- Most frequent incident types per area
- Peak emergency hours

This helps barangay and municipal authorities such as the Philippine National Police and the Bureau of Fire Protection allocate resources efficiently, plan patrol schedules, and implement preventive programs.

